GOOGLE CLOUD SKILLS | HANDS-ON LAB REPORT

Create and Configure VPC Networks

Associate Cloud Engineer path · Lab ID: CBL057

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PLATFORM	Google Cloud (Compute Engine, VPC)
LEVEL / DURATION	Introductory / 45 minutes
FINAL SCORE	100 / 100

1. Overview

A Google Cloud VPC network is a global, virtualized network that provides networking for Compute Engine VM instances, Kubernetes Engine containers and App Engine flexible environments. A VPC is a global resource made up of regional subnets connected by Google's global network, and VPC networks are logically isolated from each other. Without a VPC network, no VM instance can be created.

In this lab I created an auto mode VPC network with firewall rules and two VM instances, converted it to a custom mode network, created two more custom mode networks, and tested connectivity between VM instances across the networks.

2. Objectives

- Explore the default VPC network (subnets, routes, firewall rules)
- Create an auto mode network with firewall rules
- Convert an auto mode network to a custom mode network
- Create custom mode VPC networks with firewall rules
- Create VM instances using Compute Engine
- Explore connectivity for VM instances across VPC networks

3. Resources Built

Networks and subnets

Network	Mode	Subnet	Region	IPv4 range
mynetwork	Auto → Custom	Auto-created subnets	us-central1, asia-east1	Auto-assigned (10.128.0.0/9 block)
managementnet	Custom	managementsubnet-us	us-central1	10.240.0.0/20
privatenet	Custom	privatesubnet-us	us-central1	172.16.0.0/24
privatenet	Custom	privatesubnet-notus	asia-east1	172.20.0.0/20

VM instances

VM	Network	Zone	Internal IP	External IP
mynet-us-vm	mynetwork	us-central1-a	10.128.0.2	34.70.58.19
mynet-notus-vm	mynetwork	asia-east1-b	10.140.0.2	34.80.118.180
managementnet-us-vm	managementnet	us-central1-a	10.240.0.2	34.170.41.255
privatenet-us-vm	privatenet	us-central1-a	172.16.0.2	34.172.180.167

External IPs are ephemeral and were released when the lab ended.

4. Step-by-Step Walkthrough

Task 1 – Explore the default network

The default network has a subnet in each region, a route for each subnet plus a default internet gateway route (0.0.0.0/0), and four ingress firewall rules: default-allow-icmp, default-allow-rdp, default-allow-ssh and default-allow-internal. I deleted the firewall rules and then the default network. With no VPC network there were no routes and no firewall rules, and creating a VM instance failed with the error *"No more networks available"*.

Task 2 – Create an auto mode network

I enabled the required APIs, then created **mynetwork** in Automatic subnet mode with all the standard firewall rules (allow-custom, allow-icmp, allow-rdp, allow-ssh).

```
gcloud services enable \
  iap.googleapis.com \
  networkmanagement.googleapis.com
```

I added an IAP firewall rule so SSH could be tunnelled through Identity-Aware Proxy:

Setting	Value
Name	allow-iap-ssh
Network	mynetwork
Direction / Action	Ingress / Allow
Priority	1000
Targets	Specified target tags: iap-gce
Source IPv4 range	35.235.240.0/20
Protocol / port	tcp:22

I created **mynet-us-vm** (us-central1-a) and **mynet-notus-vm** (asia-east1-b), both e2-medium running Debian GNU/Linux 12 with the network tag *iap-gce*. I connected with:

```
gcloud compute ssh mynet-us-vm \
  --zone=us-central1-a \
  --tunnel-through-iap
```

Finally I converted mynetwork to **Custom** subnet mode so that new subnets are not created automatically as new regions appear, which could overlap with manually planned IP ranges. Custom mode is recommended for production.

Task 3 – Create custom mode networks

managementnet was created in the console. **privatenet** was created with gcloud:

```
gcloud compute networks create privatenet --subnet-mode=custom

gcloud compute networks subnets create privatesubnet-us --network=privatenet \
  --region=us-central1 --range=172.16.0.0/24

gcloud compute networks subnets create privatesubnet-notus --network=privatenet \
  --region=asia-east1 --range=172.20.0.0/20
```

Firewall rules for both networks allow ICMP, SSH (tcp:22) and RDP (tcp:3389) from 0.0.0.0/0:

```
gcloud compute firewall-rules create privatenet-allow-icmp-ssh-rdp \
  --direction=INGRESS --priority=1000 --network=privatenet \
  --action=ALLOW --rules=icmp,tcp:22,tcp:3389 --source-ranges=0.0.0.0/0
```

Then I created **managementnet-us-vm** (console, e2-micro, managementsubnet-us) and **privatenet-us-vm** (gcloud):

```
gcloud compute instances create privatenet-us-vm --zone=us-central1-a \
  --machine-type=e2-micro --subnet=privatesubnet-us \
  --image-family=debian-12 --image-project=debian-cloud \
  --boot-disk-size=10GB --boot-disk-type=pd-standard \
  --boot-disk-device-name=privatenet-us-vm
```

The console confirms the result. All three networks are in Custom mode, and the firewall rules and VM instances are in place:

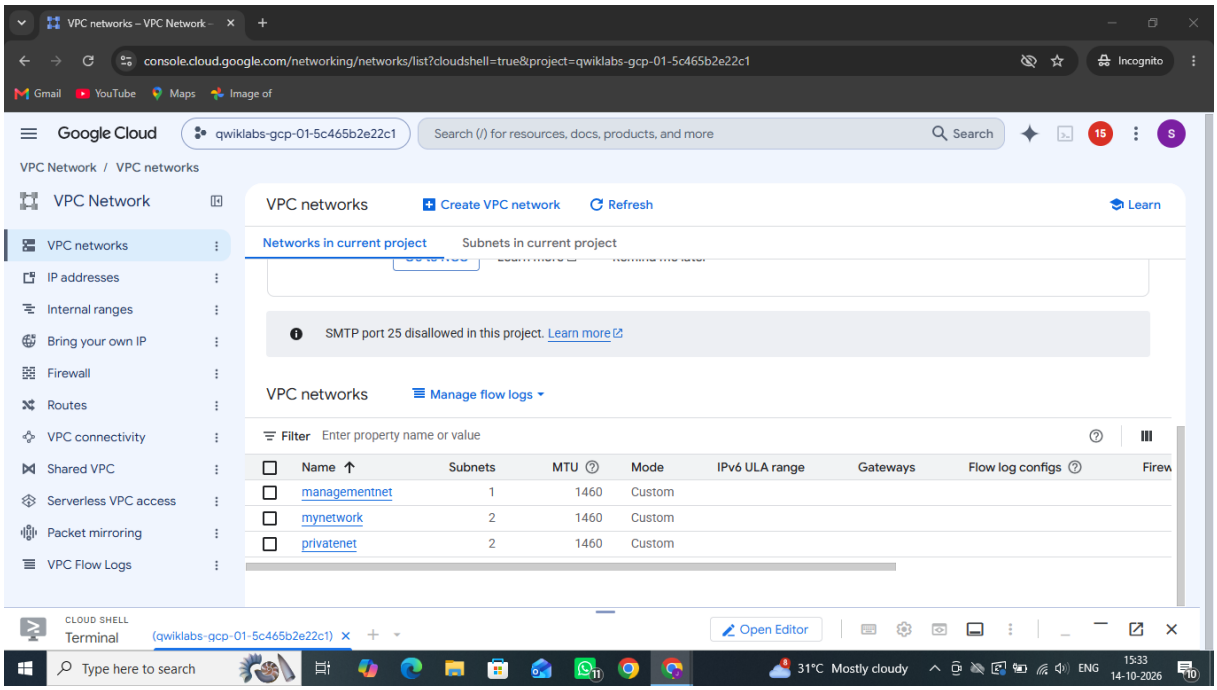


Figure 1: VPC networks – managementnet (1 subnet), mynetwork (2 subnets) and privatenet (2 subnets), all in Custom mode.

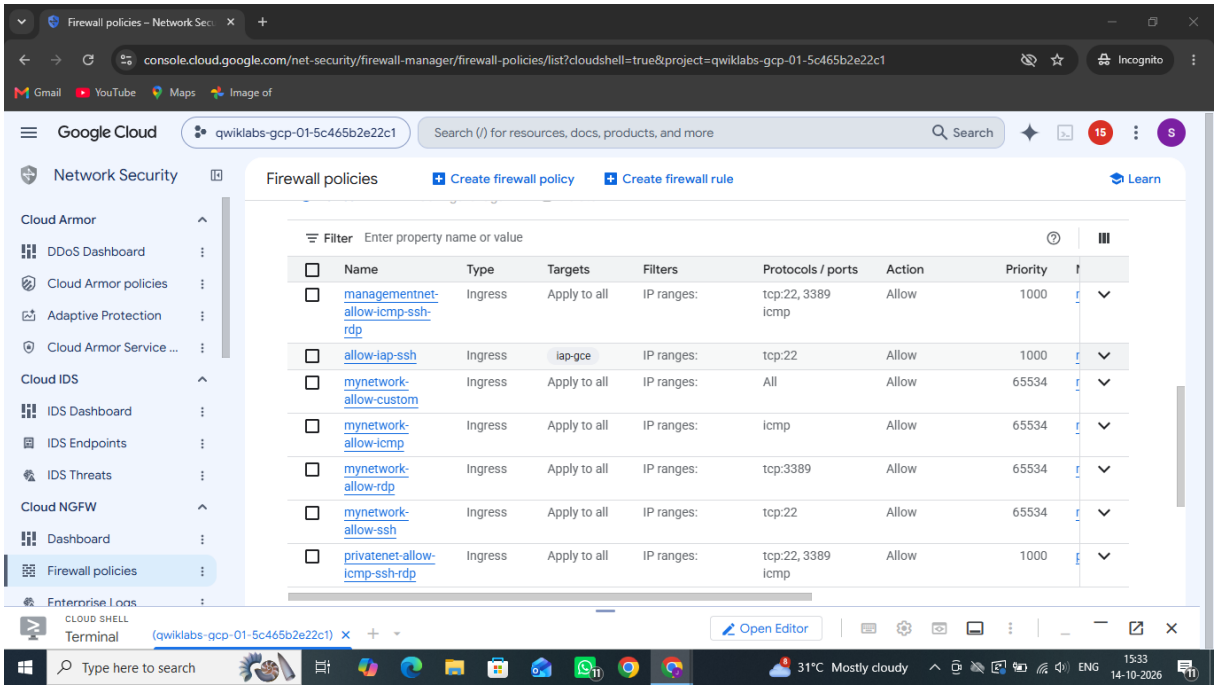


Figure 2: Firewall rules – allow-iap-ssh (target tag iap-gce), the mynetwork rules, managementnet-allow-icmp-ssh-rdp and privatenet-allow-icmp-ssh-rdp.

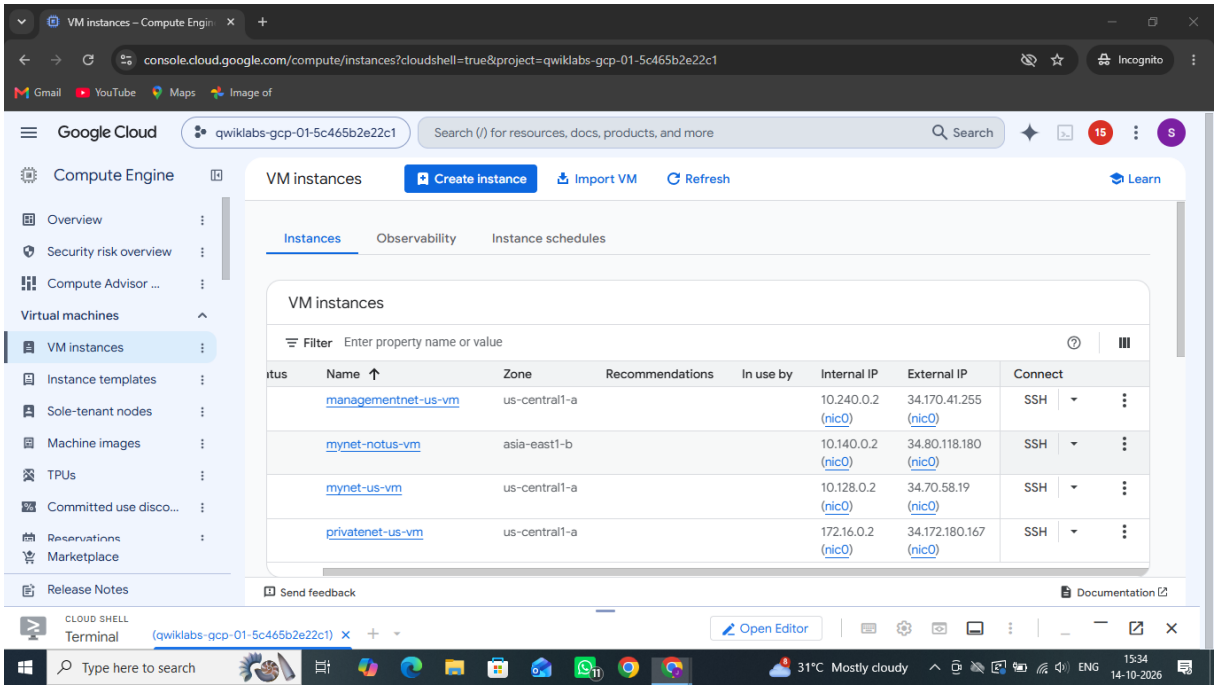


Figure 3: VM instances with their zones, internal IPs and external IPs. Three instances are in us-central1-a and one is in asia-east1-b, spread across three VPC networks.

Task 4 – Explore connectivity across networks

I connected to **mynet-us-vm** over IAP SSH and ran ping tests against the external and internal IP addresses of the other instances.

External IPs. Pings to external addresses succeeded, including to instances in another region and other VPC networks. Public reachability is controlled only by the ICMP firewall rules.

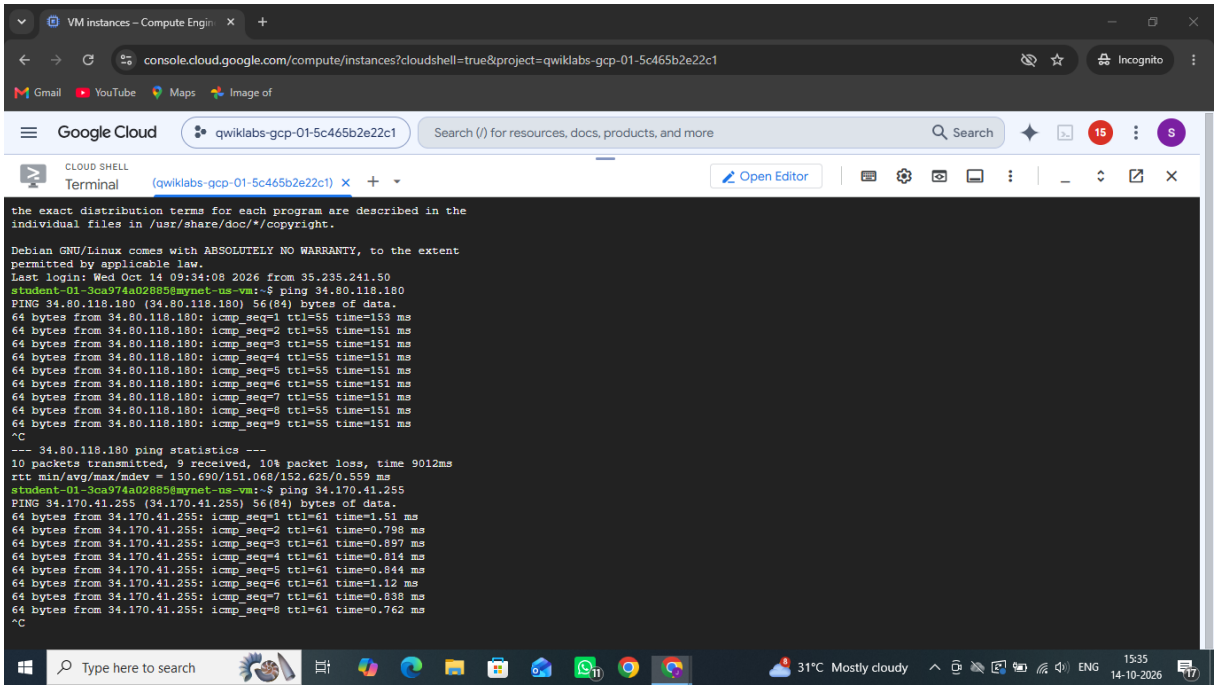


Figure 4: SSH session on mynet-us-vm. Ping to mynet-notus-vm (34.80.118.180, asia-east1) succeeded with about 151 ms latency (9 of 10 replies received before stopping), then ping to managementnet-us-vm (34.170.41.255) started.

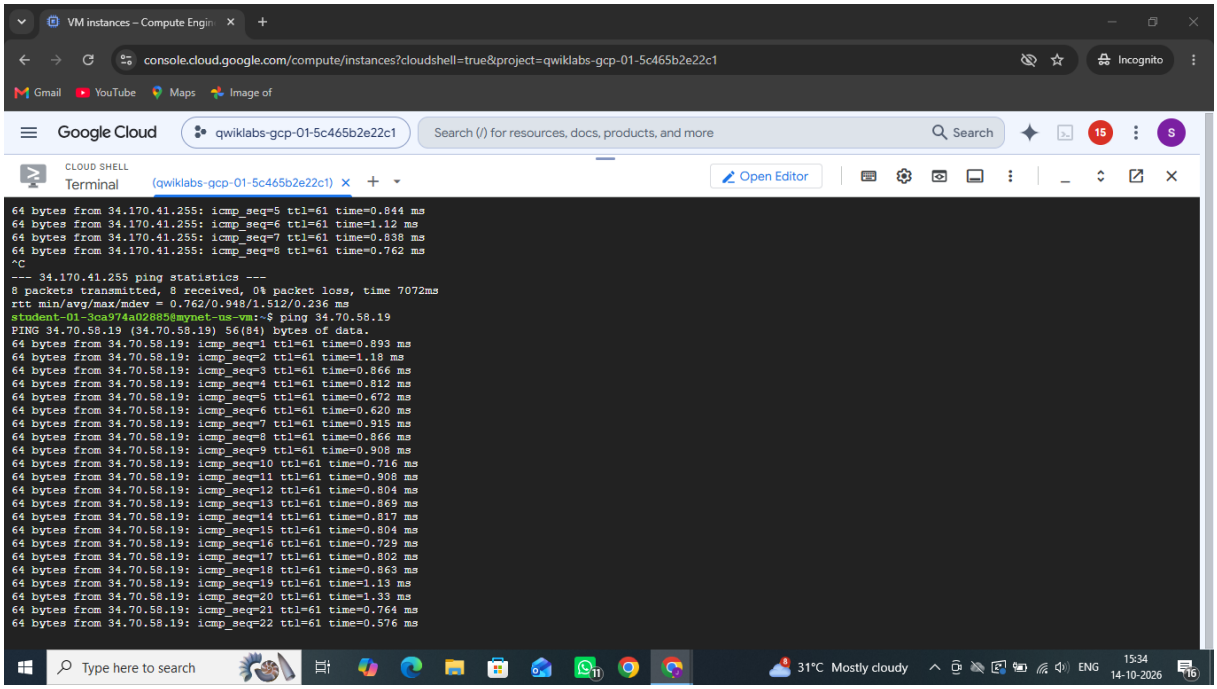


Figure 5: Ping to managementnet-us-vm external IP (8 sent, 8 received, 0% loss), then ping to 34.70.58.19 (mynet-us-vm's own external IP) with sub-millisecond replies.

Internal IPs. Pings to internal addresses worked only inside the same VPC network, even across regions.

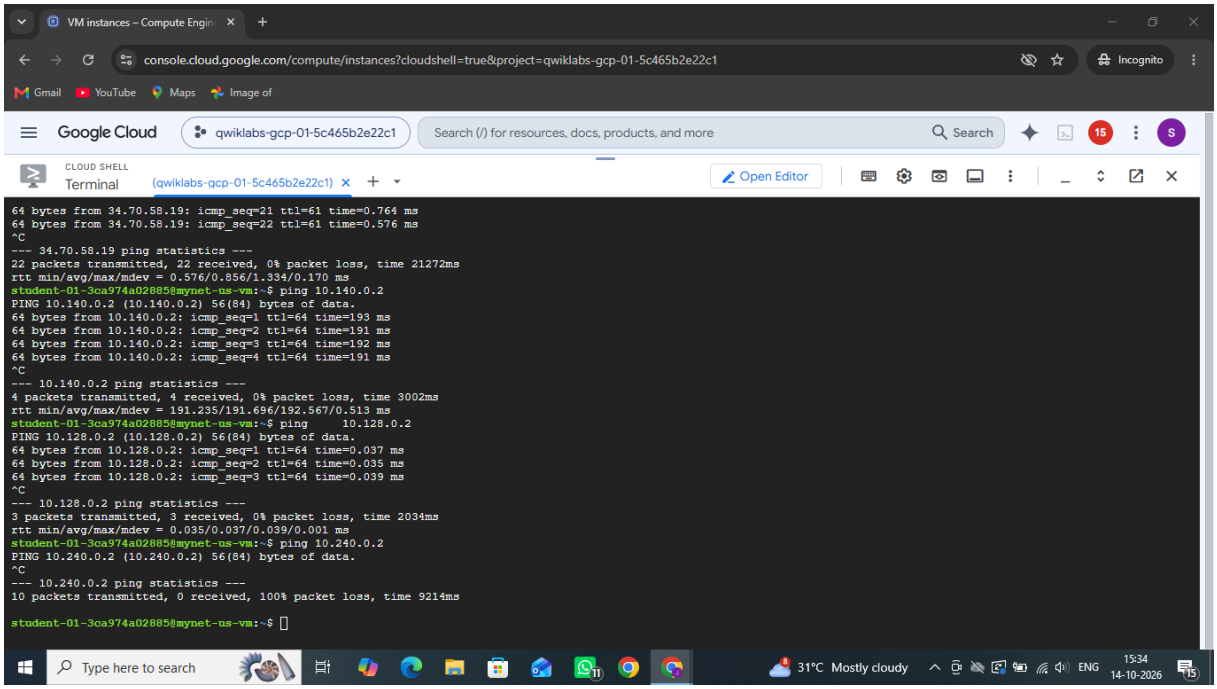


Figure 6: Internal pings from mynet-us-vm. 10.140.0.2 (mynet-notus-vm, same network, different region) answers with 0% loss; 10.128.0.2 (itself) answers with 0% loss; 10.240.0.2 (managementnet-us-vm, different network, same zone) shows 100% packet loss.

5. Connectivity Results

From	To	Address	Result	Why
mynet-us-vm	mynet-notus-vm	External 34.80.118.180	Reachable (about 151 ms)	ICMP allowed by mynetwork-allow-icmp
mynet-us-vm	managementnet-us-vm	External 34.170.41.255	Reachable (0% loss)	ICMP allowed by managementnet-allow-icmp-ssh-rdp
mynet-us-vm	mynet-notus-vm	Internal 10.140.0.2	Reachable (about 191 ms)	Same VPC network, across regions
mynet-us-vm	managementnet-us-vm	Internal 10.240.0.2	**Blocked (100% loss)**	Different VPC network, isolated by default

Knowledge-check answers

- **Without a VPC network there are no routes, and you cannot create VM instances, containers or App Engine apps:** True.
- **Firewall rule that allows ping to mynet-notus-vm's external IP:** mynetwork-allow-icmp.
- **Instances pingable from mynet-us-vm by internal IP:** only mynet-notus-vm, because it is the only other instance on the same VPC network.

6. Key Takeaways

- VPC networks are global, while subnets are regional. Instances in the same VPC can talk over internal IPs across regions.
- Being in the same zone does not give connectivity. Being in the same VPC network does.

- VPC networks are isolated by default. Internal communication between networks needs VPC peering or VPN.
- Auto mode networks suit prototyping. Custom mode gives control over regions and non-overlapping IP ranges, and is recommended for production.
- Every network has implied rules that deny all ingress and allow all egress. Explicit firewall rules open specific traffic.
- IAP TCP forwarding (source range 35.235.240.0/20) allows SSH to VMs without exposing port 22 to the internet.
- The same network, subnet, firewall and VM resources can be created with the console or with gcloud.

7. Conclusion

The lab was completed with a score of 100/100. I built three VPC networks, applied firewall rules, deployed four VM instances and verified that traffic flows within a VPC network but is isolated between networks, a foundation for later work on VPC peering, VPN and hybrid connectivity.